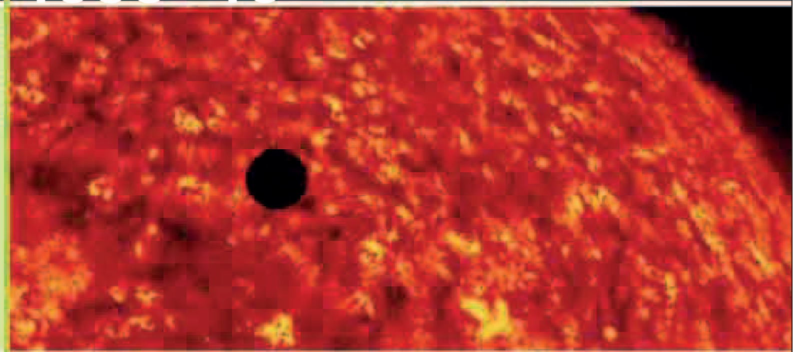
René Descartes asserted that knowledge had to be distinct and precise.

> **EACH PROBLEM THAT I SOLVED BECAME A RULE** WHICH SERVED AFTERWARD TO SOLVE **OTHER PROBLEMS.**

René Descartes, French philosopher and mathematician, from *Discourse on Method*, 1637



In 1639, Jeremiah Horrocks was the first person known to record a transit of Venus as the planet moved across the surface of the Sun.

IN 1636, THE FRENCH MATHEMATICIAN MARIN MERSENNE published a treatise on the **mathematical analysis of musical sound**, in which he described laws to explain a stretched string's **frequency of vibration**. He stated that the frequency was lower in longer strings but increased with more stretching force.

Following Italian physicist Galileo Galilei's conviction for heresy in 1633, French philosopher and mathematician **René Descartes** delayed the release of *The World*—a bold account of his scientific views, including an agreement with Galileo's theory of Earth revolving around the Sun. Part of the text appeared in 1637, in *Discourse on the Method of Properly Conducting One's Reason and of Seeking the Truth in the Sciences*, which included essays on astronomy, geometry, and optics. In the book's appendix, *Geometry*, Descartes explained how **algebra and geometry were connected**. Two quantities, x and y, could be represented on two intersecting coordinate lines, the x and y axes, in a graph, and the relationship between the two could be represented in an algebraic equation. Another French mathematician, **Pierre de Fermat** (1601–65), had independently devised this method in 1629, but it was given Descartes' name, and was called the **Cartesian coordinate system** (see panel, below).

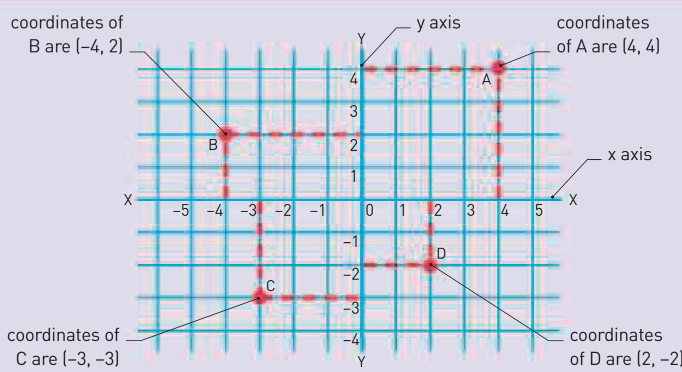
Fermat became better known for his **"Last Theorem,"** which stated that no positive whole numbers fit the equation $a^n + b^n = c^n$, where n is greater than 2. He wrote the theorem in the margin of an old textbook, claiming that he had proof for the theorem but no room to write it. Independent proof for the theorem was not found until 1995.

PIERRE DE FERMAT (1601–65)

Trained as a lawyer, Pierre de Fermat influenced several branches of mathematics. While he considered himself an amateur, and often refrained from providing proof for his discoveries, his work in geometry anticipated that of René Descartes. In 1654, Fermat corresponded with Blaise Pascal and helped develop probability theory.

CARTESIAN COORDINATE SYSTEM

When Descartes and Fermat realized how coordinates link algebra and geometry, it was a major breakthrough for science. Coordinates consist of two intersecting axis lines—x (horizontal) and y (vertical). It is possible to define the position of any point within this two-dimensional space by stating the values of x and y, that is, the x and y coordinates.



IN 1638, GALILEO GALILEI published his final word on physics: *Discourses and Mathematical Demonstrations Relating to Two New Sciences*, which dealt with the **strength of materials and kinematics**—the study of the motion of bodies without reference to mass or force. The Inquisition had banned publication of any work by Galileo after his trial in 1633. However, *Discourses* was published in Leiden, Netherlands, where the Inquisition had little influence.

English astronomer **Jeremiah Horrocks** (1618–41) had been studying Venus and estimated that the planet would pass the Sun on December 4, 1639. His prediction was based on the understanding that transits of Venus occur in pairs, eight years apart, and each paired transit occurs more than 100 years apart. When the time came, Horrocks focused the Sun's image onto paper and spotted the shadow of Venus only 15 minutes later than his prediction. Horrocks went on to calculate **Venus's size and distance** more accurately than ever before.

In 1640, sixteen-year-old French prodigy **Blaise Pascal** (1623–62) published *Essay on Conics*, in which he described the geometric relationship that occurs when a hexagon is drawn within a circle. In doing so he completed a mathematical theorem so advanced that at first many, including René Descartes, refused to believe that the young mathematician had done it.

In 1640, English botanist **John Parkinson** (1566–1650) published a plant catalog called *Theatrum Botanicum* (*Theater of Plants*). This was the most comprehensive work of its kind at the time, and remained a popular guide for many years.

7,522 MILES
THE **DIAMETER** OF **VENUS**

May 15, 1635 France's principal botanic garden, the **Jardin du Roi**, is founded in Paris

1636 French mathematician Marin Mersenne describes his **laws of oscillation**

1637 French mathematician Pierre de Fermat claims to have solved his **"Last Theorem"**

1637 French mathematician **René Descartes** publishes *Discourse on Method, the Meteors, Dioptrics, and Geometry*

1637 American author Thomas Morton writes about **North American flora and fauna** in *The New English Canaan*

1637 Chinese encyclopedist Song Yingxing publishes a **scientific encyclopedia**, *The Workings of Heaven Revealed*

1638 Galileo publishes his **last book on physics**

December 4, 1639 Jeremiah Horrocks observes the **transit of Venus**

February 1640 French mathematician Blaise Pascal solves the "mystic hexagram" within a circle **(Pascal's theorem)**

1640 John Parkinson publishes *Theatrum Botanicum (Theater of Plants)*

October 18, 1640 Pierre de Fermat claims he has proof for his **"Little Theorem"**